

Dashboard Creation

Objective:

The purpose of the dashboard is to transform raw data and complex queries into an intuitive, interactive, and visually engaging format to support decision-making by stakeholders. It focuses on uncovering insights from customer behavior, sales performance, customer feedback, and business health.

Tools Used

- **Power BI** – for dashboard development, interactive visuals, and cross-filtering
 - **Google BigQuery (GCP)** – as the connected data source, using SQL for data querying
 - **Google Cloud Storage (GCS)** – for centralized dataset storage
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Design Principles

- **Simplicity & Clarity:** Focused on user-friendly layout and clean visual design.
- **Segmentation:** Dashboard divided into multiple pages based on analysis themes:
 1. **Customer Behavior**
 2. **Customer Feedback**
 3. **Sales Insights**
 4. **Business Questions / Strategic KPIs**
- **Interactivity:** Includes slicers (filters) by date, package, channel, customer type, etc.
- **Drilldowns & Tooltips:** Enhanced user interaction with hover-over stats, decomposable charts, and drill-throughs to details.

Pages Overview

1. Customer Behavior Dashboard

- **Key Visuals:**
 - Stacked Bar Chart: Customer type vs. gender
 - Card: Average customer age
 - Heatmap: Preferred package by customer type
 - Donut Chart: New vs. Repeat customers
 - Column Chart: Booking lead time by package type
- **Advanced Features:**
 - Custom age group segmentation
 - Filtering by booking channels and travel months

2. Customer Feedback Dashboard

- **Key Visuals:**
 - Bar Chart: Avg satisfaction by package
 - KPI Card: Top-rated travel guide
 - Dot Plot: Guide rating vs. number of reviews
 - Donut Chart: Would recommend % breakdown
 - Treemap: Refund amount by reason
- **Advanced Features:**
 - Drill-through: From package to individual feedback records
 - Gauge Chart: Promoter % against target

3. Sales Insights Dashboard

- **Key Visuals:**
 - Column Chart: Revenue by sales channel
 - Stacked Column: Gross vs. Net revenue by discount status
 - Line Chart: Revenue by travel month
 - Cards: Peak season booking metrics
- **Advanced Features:**
 - Scenario modeling buttons (What-if analysis)
 - Dynamic filters based on seasonal booking behavior

4. Business Questions Dashboard

- **Key Visuals:**
 - Scatter Plot: Satisfaction vs. Refund Rate (by package)
 - Column Chart: Refund rate by package
 - Decomposition Tree: Refund breakdown (status → reason → package)
 - Matrix Table: Customer channel preferences by age group
 - **Advanced Features:**
 - Bookmark navigation
 - Custom tooltips on financial impact visuals
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Data Connectivity

- Connected directly to **Google BigQuery** using the **ODBC connector** in Power BI.
 - Utilized **Import mode** to optimize performance and responsiveness due to stable dataset size.
 - Scheduled refresh and manual sync options available.
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Result

The final dashboards offer:

- **Real-time visibility** into business metrics
- **Actionable insights** for campaign planning and customer engagement
- A foundation for continuous improvement and stakeholder alignment